
Results

Our comprehensive evaluation of instruction-following capabilities encompassed all 331 models available via OpenRouter as of October 14, 2025. We verified all 331 models for basic functionality, of which 256 passed verification and were subsequently evaluated using our twenty diagnostic prompts. Following our two-stage verification protocol, we first assessed basic endpoint functionality and subsequently evaluated all verified working models across the twenty diagnostic tests.

Model Verification Results

Of the 331 models initially retrieved from OpenRouter, 256 models (77.3%) successfully completed the basic verification test with the correct answer "Paris", demonstrating both functional API endpoints and coherent response generation. The remaining 75 models (22.7%) either failed verification due to endpoint errors, timeout issues, rate limiting on free tiers, or provided incorrect responses. This verification step proved essential: it prevented potential confounding from non-functional endpoints that could obscure genuine instruction-following patterns and ensured all tested models had confirmed operational API access.

The evaluation of all 256 verified models on 20 diagnostic tests resulted in 5,120 individual test evaluations, achieving a 99.4% API success rate, which validates the robustness of our testing methodology.

Instruction-Following Performance by Task

Figure 1 presents the overall performance matrix across all verified models and twenty diagnostic tasks. Task-specific success rates varied substantially, ranging from 2.7% for Test 5 (String Manipulation Chain) to 80.1% for Test 13 (Selective Text Processing), with an overall pass rate of 43.7% across all model-task combinations.

The empirical task difficulty ranking revealed counterintuitive patterns. Test 4 (Prime After 10000), requiring numerical identification, achieved a 66.4% success rate, substantially higher than Test 1 (Multi-step String Manipulation), which achieved only 7.8% despite requiring operations typically mastered in elementary education. Most notably, Test 5 (String Manipulation Chain) proved to be the hardest test with only a 2.7% pass rate, while Test 13 (Selective Text Processing) was the easiest at 80.1%. This dissociation between computational complexity and instruction-following success rates, where simple string operations prove more challenging than mathematical computations, warrants further investigation (see Discussion).

Performance by Instruction Category

We analyzed performance patterns by grouping tasks according to their primary instruction-following requirement: string manipulation, mathematical operations, data processing, format conversion, and constraint compliance. Figure 2 displays the comprehensive difficulty ranking across all tests.

Analysis of category-level performance revealed striking differences (Table 3). String manipulation tasks achieved only 12.0% average success rate, while constraint compliance tests reached 66.9%. Mathematical operations (44.9%) and data processing (45.5%) showed intermediate difficulty, with format conversion tasks at 57.8%. This categorical pattern demonstrates that instruction-following difficulty is not primarily determined by computational complexity but rather by the nature of the required operation. Notably, simple string operations requiring precise character-level

manipulation proved far more challenging than complex mathematical computations, despite the latter involving cognitively more demanding operations.

Model Family Comparisons

Performance patterns varied substantially across model provider families (Figure 3, Table 4). Among providers with at least 3 models, x-ai models achieved the highest mean success rate (79.3% across 7 models), followed by sao10k (57.0%, 5 models) and Anthropic (56.8%, 11 models). Major commercial providers showed more modest performance: Google models averaged 52.1% (21 models), while both OpenAI (48.3%, 36 models) and Qwen (48.1%, 32 models) hovered around the overall average of 43.7%. Lower-tier providers like Microsoft (37.9%, 7 models), Baidu (33.8%, 4 models), and Nvidia (33.3%, 3 models) demonstrated weaker instruction-following capabilities. Notably, the top individual model (qwen/qwen-plus-2025-07-28:thinking with 100% pass rate) came from Qwen despite the provider’s average performance, illustrating substantial within-family variation and suggesting that provider identity alone does not determine instruction-following capability.

The substantial provider-level variation (ranging from 33.3% to 79.3% among providers with ≥ 3 models) indicates that training methodologies, architectural choices, and optimization objectives significantly impact instruction-following capabilities. Interestingly, performance patterns across task categories remained relatively consistent across providers, suggesting that fundamental challenges in instruction following (particularly string manipulation) affect models universally regardless of their training approach.

Top Performing Models

Table 1 presents the top 15 performing models from our comprehensive evaluation. The best model, qwen/qwen-plus-2025-07-28:thinking, achieved a perfect 100% pass rate across all 20 tests. Four models from x-ai (grok-4-fast, grok-code-fast-1, grok-4, and grok-3-mini) demonstrated exceptional performance (90-95%), highlighting the provider’s strong instruction-following capabilities. Multiple models from the 75% tier (including several from Qwen, OpenAI, Anthropic, and x-ai) indicate a performance plateau where models handle most but not all instruction types consistently. Notably, only one model achieved perfect performance, while ten models scored 0%, demonstrating the wide spectrum of instruction-following capabilities across the contemporary LLM landscape.

Discussion

Our evaluation of instruction-following capabilities, conducted through comprehensive verification of all 331 models available via OpenRouter (October 14, 2025) followed by systematic assessment of 256 verified working models using twenty diagnostic prompts, reveals significant insights into the current state of instruction adherence in contemporary LLMs. We hypothesized that: (1) instruction-following capabilities would vary significantly across contemporary language models, (2) different types of instructions would pose varying levels of difficulty, (3) model size would not be the sole determinant of instruction-following performance, and (4) provider-specific training approaches would result in distinct performance patterns. The results demonstrate substantial variability in models’ abilities to follow precise instructions, with failure patterns that transcend model size, architecture, and training methodology, largely confirming our hypotheses. Notably, our verification protocol identified that 22.7% (75

Table 1. Top 15 Performing Models on 20 Diagnostic Tests

Rank	Model	Pass Rate
1	qwen/qwen-plus-2025-07-28:thinking	100.0%
2	x-ai/grok-4-fast	95.0%
3	x-ai/grok-code-fast-1	90.0%
4	x-ai/grok-4	90.0%
5	openai/gpt-oss-20b:free	85.0%
6	openai/gpt-oss-120b	85.0%
7	qwen/qwen3-vl-235b-a22b-instruct	75.0%
8	perplexity/sonar-reasoning-pro	75.0%
9	openai/gpt-oss-20b	75.0%
10	tencent/hunyuan-a13b-instruct	75.0%
11	openai/gpt-5-codex	75.0%
12	anthropic/claude-3.7-sonnet:thinking	75.0%
13	anthropic/claude-3.7-sonnet	75.0%
14	x-ai/grok-3-mini	75.0%
15	qwen/qwen3-max	75.0%

Table 2. Test Difficulty Analysis (10 Hardest Tests)

ID	Test Name	Pass Rate	Category
5	String Manipulation Chain	2.7%	string_manipulation
3	Remove Repeated Letters JSON	5.5%	data_processing
1	Multi-step String Manipulation	7.8%	string_manipulation
18	Vowel Count Sorting	7.8%	data_processing
17	Same Start/End Letter	8.6%	string_manipulation
16	Matrix Diagonal Difference	12.5%	mathematical
8	Deduplication and Position Multiply	16.8%	mathematical
20	String Replace with Newlines	28.9%	string_manipulation
14	Prime JSON	48.0%	format_conversion
11	Base64 Encoding	57.4%	format_conversion

Table 3. Performance by Test Category

Category	Pass Rate
string_manipulation	12.0%
mathematical	44.9%
data_processing	45.5%
format_conversion	57.8%
constraint_compliance	66.9%